System Design Specification: Modular Base CSE and Multi-Instance Vehicle AE Architecture

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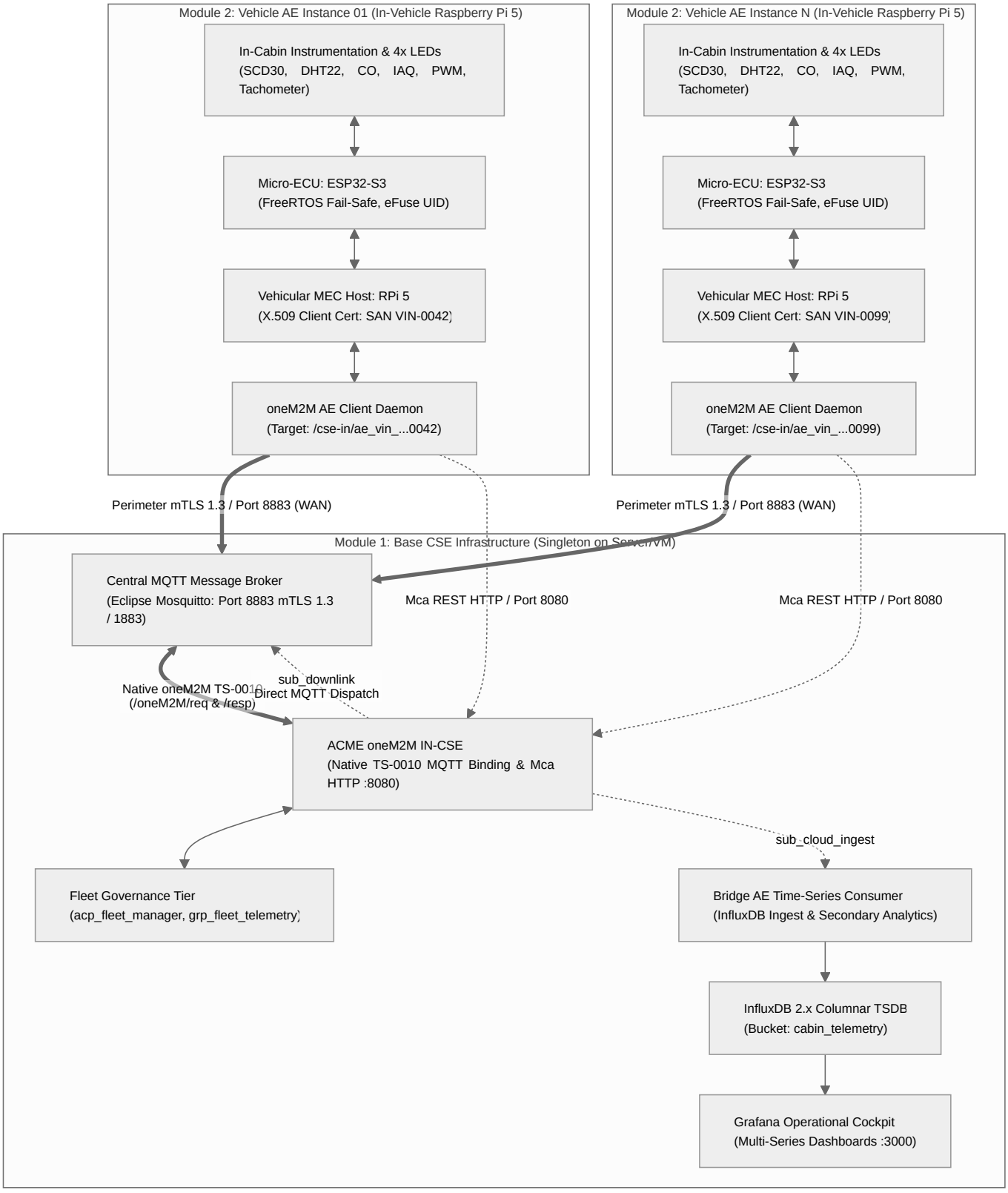
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Associated Baseline: acme_cse_gateway_capability_review.md

1. EXECUTIVE SUMMARY AND ARCHITECTURAL PRINCIPLES

This specification defines the structural partitioning, physical host allocations, interface contracts, identity-binding protocols, and deployment lifecycle for the SDV Cyber-Physical Environmental Monitoring and Closed-Loop HVAC Regulation platform.

The architecture enforces a strict demarcation between the central infrastructure platform and distributed vehicular edge compute nodes:

1. **Base CSE Infrastructure Module (Singleton Instance — Server / Virtual Machine):** * **Host Platform:** Deployed on an external connected device—a **central physical server, cloud instance, or virtual machine (VM)**. * **Role:** Serves as the authoritative Common Services Entity (ACME oneM2M IN-CSE), central MQTT message broker (Eclipse Mosquitto terminating perimeter mTLS 1.3), fleet governance policy repository, time-series data lake (InfluxDB 2.x), and fleet-wide operational cockpit (Grafana Enterprise). * **Cardinality: Strictly a single, authoritative instance per operational cluster ($C = 1$).**
2. **Vehicle Application Entity (AE) Module (Multi-Instance — In-Vehicle Raspberry Pi 5):** * **Host Platform:** Deployed **directly onboard each physical vehicle on a dedicated Raspberry Pi 5** single-board computer (vehicular MEC host). * **Role:** Interfaced directly with in-cabin physical instrumentation (ESP32-S3 micro-ECU, Sensirion SCD30, DHT22, CO, IAQ/VOC, AQI transducers, 4-channel LED actuator matrix, WS2812B annunciator, and closed-loop tachometer). The Raspberry Pi 5 executes the uniquely identified Application Entity (AE) client daemon, local sensor ingress/pre-processing, mTLS client credentials, and downlink actuation dispatcher. * **Cardinality: Deployable across N uniquely identified, concurrent, and independent vehicular instances ($i \in \{1, 2, \dots, N\}$).**



2. CONSOLIDATION OF VEHICULAR EDGE GATEWAY FUNCTIONS INTO ACME ONEM2M CSE

As established in the architectural capability review (acme_cse_gateway_capability_review.md), the platform consolidates legacy edge gateway microservices directly into native ETSI TS 118 101 middleware capabilities.

2.1 Excised Microservices and Absorbed Capabilities

1. **Decommissioning of Node-RED Ingress Proxy (`nodered_ipe`):** * *Legacy Role:* Acted as an intermediate Node.js container translating raw MQTT topics into HTTP REST `<m2m:cin>` instances, and translating HTTP webhooks into downlink MQTT actuation commands. * *Consolidated Architecture:* Completely decommissioned and excised from the deployment mesh. ACME oneM2M IN-CSE natively terminates and dispatches oneM2M primitives over MQTT via **oneM2M TS-0010 MQTT protocol bindings** (`[mqtt]` in `deploy/onem2m/acme.ini`). The vehicle node publishes ContentInstances directly to `/oneM2M/req/...` and receives actuation

downlinks on `/oneM2M/req/id-in/{aei}/json`. * *Performance Gain:* Reclaims ~220 MB RAM and reduces closed-loop actuation propagation latency by $40\text{--}80\,\text{ms}$.

2. **In-CSE Multi-Hazard Health Classification:** * *Legacy Role:* The standalone Python `bridge_ae` microservice evaluated all sensor divergence, stale timestamps, and motor stalls. * *Consolidated Architecture:* ACME CSE's internal scripting engine (`acme-scripts` with `onCreate` hooks) absorbs multi-hazard rule evaluation directly upon ContentInstance creation. The CSE tags `<cin>` resources with semantic labels (`lbl: ["health:FRESH", "delta_t:0.32", "delta_rh:1.8", "stall:false"]`) and raises emergency alarms internally. `bridge_ae` is streamlined into a lightweight, non-blocking time-series persistence consumer.
3. **Edge Time-Series Buffering and Storage Caps:** * *Consolidated Architecture:* Rather than running a heavyweight edge TSDB for short-term buffering, ACME CSE enforces standard oneM2M container lifecycle bounds:
   - `mni: 1440`: Buffers 1,440 historical instances (24 hours at 1-minute cadence).
   - `mbs: 2097152`: Strict 2 MB memory-footprint cap protecting vehicular storage.
   - `mia: 86400`: 24-hour automatic TTL instance pruning.
   - `missingDataDetect: true`: Built-in sensor loss detection without custom watchdog daemons.
4. **Dynamic Fleet Group Fan-Out (`<grp>`, `<fopt>`):** * *Consolidated Architecture:* Centralised group aggregation allows atomic fleet status queries (`/grp_fleet_telemetry/fopt/la`) and multi-vehicle broadcast actuations without external orchestrator scripts.
5. **Dynamic Semantic Policy Engine (`<acp>`):** * *Consolidated Architecture:* Replaces static broker ACL text files with standard oneM2M Access Control Policies, dynamically binding privileges to verified AE-IDs.

2.2 Architectural Segregation: Retained Dedicated Platforms

To guarantee cyber-physical safety, cryptographic robustness, and high-performance analytics, specific functions are strictly retained on dedicated platforms outside the CSE:

Subsystem	Placement	Architectural Rationale
Mutual TLS 1.3 Termination	**Eclipse Mosquitto (Port 8883)**	Cryptographic perimeter security, hardware ECDSA P-256 acceleration, and X.509 client certificate verification must terminate at a hardened C broker, protecting the Python ASGI/WSGI stack.
Long-Term Multi-Series TSDB	**InfluxDB 2.x Columnar Engine**	High-cardinality nanosecond time-series analytics and Flux queries across months of fleet operation exceed the capacity of oneM2M SQLite/TinyDB metadata backends.
Operational Cockpit	**Grafana Enterprise (Port 3000)**	Fleet-wide visual observability, multi-series dashboards, dynamic alerting, and operator control cockpits.
Autonomous Safety Overrides	**ESP32-S3 FreeRTOS Firmware**	Safety-critical fail-safe loops (e.g. 50% baseline purge ventilation during communication loss or CO emergency) execute deterministically in microcontroller firmware independent of edge host or network state.

3. MODULE 1: BASE CSE INFRASTRUCTURE SPECIFICATION (SINGLETON ON SERVER / VIRTUAL MACHINE)

3.1 Host Platform and Deployment Topology

- **Host Platform:** Dedicated physical server, cloud instance, or virtual machine (VM) (Linux x86_64 or aarch64 server environment).
- **Singleton Constraint:** Exactly one Base CSE infrastructure instance exists per operational cluster. It operates with a stable, routable network identity (e.g. `GATEWAY_IP=192.168.122.91` or cluster FQDN).
- **Deployment Packaging:** Packaged and deployed via `deploy/docker-compose.yml` on the central host.

3.2 Core Component Responsibilities

1. **ACME oneM2M IN-CSE (`sdv_onem2m_cse`):** * Exposes standard Mca HTTP(S) reference point on port `8080` and native oneM2M TS-0010 MQTT binding on internal port `1883`. * Manages the root `<m2m:cb>` (CSEBase) identified by `cse_id` (`id-in`) and accessible at path `/{{ cse_rn }}` (`/cse-in` or `/sdv-cse`). * Enforces semantic schema validation, resource lifecycles, and access control policies.
2. **Fleet Governance Layer (Pre-Provisioned Hierarchy):** * **Fleet Access Control Policy (`acp_fleet_manager`):** Establishes default administrative and onboarding privileges. * **Fleet Fan-Out Group (`grp_fleet_telemetry`):** An aggregate `<m2m:grp>` whose member IDs (`mid`) dynamically index the raw telemetry containers of all onboarded vehicles. Supports atomic fleet queries via `/grp_fleet_telemetry/fopt/la`. * **Downlink & Cloud Ingestion Subscriptions:** Dispatches asynchronous change events to subscriber targets.
3. **Data Health & Storage Pipeline:** * **Central Mosquitto Broker (`sdv_mosquitto_broker`):** Terminates network mTLS 1.3 on port 8883 from all vehicular Raspberry Pi 5 nodes. * **Bridge AE Time-Series Consumer (`sdv_bridge_ae`):** Consumes `<subscription>` notifications and writes clean records to InfluxDB. * **InfluxDB 2.x TSDB (`sdv_influxdb`):** Historical time-series persistence. * **Grafana Operational Cockpit (`sdv_grafana`):** Observability and fleet-wide KPI visualisation.

3.3 Initial State Guarantee

Upon initial bootstrap, the Base CSE module **must contain zero vehicular Application Entities**. No vehicle node shall be present until dynamically onboarded by an authentic Vehicle AE module running on an in-vehicle Raspberry Pi 5.

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4. MODULE 2: VEHICLE AE SPECIFICATION (MULTI-INSTANCE ON IN-VEHICLE RASPBERRY PI 5)

4.1 Host Platform and Deployment Topology

- **Host Platform:** Dedicated **Raspberry Pi 5 single-board computer** installed inside each individual vehicle.
- **Multi-Instance Constraint:** For N vehicles in the fleet, N distinct Raspberry Pi 5 units are deployed. Each unit operates completely independently with zero shared local state, executing its own isolated edge runtime.
- **Network Connection to Base CSE:** Connects over vehicular wide-area networks (cellular 4G/5G, road-side Wi-Fi, or vehicular mesh) back to the central Server/VM on port 8883 (mTLS 1.3) and port 8080 (oneM2M Mca).

4.2 Physical and Edge Compute Stack

Each in-vehicle Raspberry Pi 5 orchestrates two physical and computational tiers:

Tier A: Physical Sensing & Actuation Node (ESP32-S3 Micro-ECU)

- **Hardware Interface to Raspberry Pi 5:** Directly connected via a dedicated physical serial bus (USB-CDC on `/dev/serial/by-id/usb-Espressif_ESP32-S3_...` or hardware GPIO UART) with parity checking.
- **Transducers (7 Environmental Channels):**
- Sensirion SCD30: NDIR CO_2 (0–10,000 ppm), dry-bulb temperature, relative humidity over $\mathrm{I}^2\mathrm{C}$ Fast-Mode (400 kHz) with hardware CRC-8 verification.
- DHT22: Redundant temperature and relative humidity over 1-Wire bus with parity checksum.
- Carbon Monoxide (CO) Sensor: Toxic gas detection (0–1,000 ppm) with emergency alert interrupt.
- Cabin Air Quality (IAQ/VOC): Volatile organic compound index (0–500) for passenger air quality.
- Exterior Air Quality (AQI): Ambient outdoor index for fresh-air intake vs cabin air recirculation damper control (ΔIAQ).
- **Actuators (4-Channel Emulation Matrix & Closed-Loop Tachometer):**
- Blue LED (GPIO 18): Purge blower ventilation driven at 25.0 kHz PWM with 10-bit resolution (0--100% duty cycle).
- Red LED (GPIO 17): Auxiliary cabin PTC heating core driven by digital output ($T_{\text{cabin}} < 19.0\,^{\circ}\text{C}$).
- Green LED (GPIO 16): Baseline eco-ventilation driven by digital output (nominal air quality).
- Amber LED (GPIO 15): Dehumidification and warning alert driven by digital output ($RH > 65\%$ or degraded data health).
- WS2812B RGB LED (GPIO 38): Optical state annunciator indicating network, sensor, and safety status.
- Hall-Effect Tachometer Feedback (GPIO 19): Interfaced via Pulse Counter (PCNT) Unit 0 for closed-loop blower stall detection.
- **Firmware Runtime:** FreeRTOS with strict core affinity (Core 1: Sensor/PWM loop; Core 0: mTLS/Network loop).
- **Hardware Identity:** Immutable 48-bit MAC address and 128-bit UID read from write-once eFuse registers.

Tier B: Vehicular MEC Host (Raspberry Pi 5)

- **Compute Platform:** Quad-core ARM Cortex-A76 vehicular edge host running vehicular Linux.
- **Local Ingress & Formatting Agent:** Ingests raw telemetry frames from the ESP32-S3, verifies CRC-8/parity checksums, and serialises valid oneM2M JSON content instances.
- **Cryptographic Identity Vault:** Securely stores the vehicle's unique ISO 3779 VIN, OTP board serial, and private key/X.509 client certificate for mTLS 1.3 authentication against the central server/VM.
- **oneM2M AE Client Agent:** Executes the onboarding reconciliation loop, periodic telemetry transmission to the Base CSE via native TS-0010 MQTT or Mca REST, and actuation downlink command dispatch to the ESP32-S3.

4.3 Dynamic oneM2M Resource Representation

When a vehicle's Raspberry Pi 5 registers with the Base CSE on the server/VM, it provisions a dedicated subtree under `<m2m:cb>`:

```
/cse-in (CSEBase on Server/VM)
 └── ae_vin_wauzzz_sdv_2026_0042 (Application Entity: <m2m:ae> deployed on RPi 5)
      ├── temperature-cabin (Container: <m2m:cnt>)
      ├── humidity-cabin (Container: <m2m:cnt>)
      ├── co2-cabin (Container: <m2m:cnt>)
      ├── co-cabin (Container: <m2m:cnt>)
      ├── air-quality-cabin (Container: <m2m:cnt>)
      ├── air-quality-exterior (Container: <m2m:cnt>)
      ├── raw-telemetry (Container: <m2m:cnt>)
      │    ├── cin_xxxx (<m2m:cin>)
      │    └── sub_cloud_ingest (<m2m:sub>)
      └── commands (Container: <m2m:cnt>)
           └── sub_downlink_actuation (<m2m:sub>)
```

5. MULTI-LAYER BINDING MECHANISM: UNIQUELY LINKING VEHICLE TO AE

To guarantee that each physical vehicle solely and accurately links to its designated AE instance on the Base CSE without cross-talk or collision, the 5-layer binding mechanism operates across the module boundary:

Layer	Binding Mechanism	Implementation Across Physical Hosts
1. Physical Hardware	ESP32 eFuse MAC + RPi 5 Board Serial $\rightarrow$ ISO 3779 VIN	**In-Vehicle RPi 5:** Embedded in local device tree and sensor frame headers.
2. Transport Security	X.509 Mutual TLS 1.3 with Subject Alternative Name (SAN)	**Network Boundary:** RPi 5 presents client cert with `SAN: URI:urn:sdv:vin:<VIN>`. Server/VM gateway authenticates cert against vehicular Root CA.
3. Semantic Identity	Deterministic Resource Name ($rn = \text{FNV-1a}(\text{VIN})$)	**Middleware Boundary:** RPi 5 registers/reconciles deterministic rn. Server/VM Base CSE binds allocated aei to client credential.
4. Access Control	Dedicated `<m2m:acp>` per vehicle ($acop = 63$ for aei)	**Server/VM Base CSE:** Restricts CRUD operations on `/ae_vin_<slug>/*` strictly to that vehicle's authenticated aei and $CAdmin$.
5. Downlink Routing	Targeted Point of Access (poa) & MQTT topics	**Network Boundary:** Actuator directives route strictly to `sdv/{vin}/control/cmd` or native topic `/oneM2M/req/id-in/{aei}/json`.

5.1 Deterministic Resource Name Equation

To prevent collisions and eliminate hardcoded aliases, the AE Resource Name is mathematically derived from the vehicle's unique identity string:

$$rn_{\text{AE}} = \text{ae_vin_} \parallel \text{slugify}\ (\text{VIN}) \parallel \text{_} \parallel \text{hex}\ \left(\quad FNV-1a_{32}(\text{VIN})\ \right)$$

This guarantees: 1. **Uniqueness:** Two distinct vehicles ($VIN_A \neq VIN_B$) yield distinct resource names ($rn_A \neq rn_B$). 2. **Idempotence:** A rebooting vehicle always resolves the exact same resource name, eliminating duplicate "ghost" AEs.

6. MULTI-DIMENSIONAL DATA HEALTH CLASSIFICATION ENGINE

The system evaluates incoming sensor telemetry against five quantitative health criteria:

$$\Delta T = \ \left| T_{\text{SCD30}} - T_{\text{DHT22}} \right| \ \leq 1.5\,^{\circ}\text{C}$$

$$\Delta RH = \ \left| RH_{\text{SCD30}} - RH_{\text{DHT22}} \right| \ \leq 10.0\,\%$$

$$\tau_{\text{age}} = t_{\text{eval}} - t_{\text{sample}} \leq 2.5\,\text{s}$$

$$\text{Stall Fault} \iff (\text{PWM} \geq 30\%) \wedge (\text{RPM} < 100)$$

$$\text{Toxic Gas Hazard} \iff \text{CO} \geq 30.0\,\text{ppm}$$

Categorical State Assignment

1. `FRESH`: $\tau_{\text{age}} \leq 2.5\,\text{s}$, $\Delta T \leq 1.5\,^{\circ}\text{C}$, $\Delta RH \leq 10\%$, and tachometer feedback confirmed. Full closed-loop regulation active.
2. `STALE`: $2.5\,\text{s} < \tau_{\text{age}} \leq 10.0\,\text{s}$. Network latency delay; hold-last-valid-value regulation logic engaged.
3. `DEGRADED`: $\Delta T > 1.5\,^{\circ}\text{C}$ or $\Delta RH > 10.0\%$. Transducer divergence; amber annunciator illuminated; system relies on conservative SCD30 readings.
4. `FAULT`: $\tau_{\text{age}} > 10.0\,\text{s}$, null reading, out-of-bounds reading, or blower motor stall detected. Numerical values excluded from time-series aggregates; blower defaults to 50% baseline fail-safe ventilation.

7. DEPLOYMENT AND OPERATIONAL SCENARIOS

7.1 Scenario A: Singleton Base CSE Bootstrap (Server / Virtual Machine)

1. The cloud or server infrastructure deploys `deploy/docker-compose.yml` on the Server/VM.
2. `sdv_onem2m_cse`, `sdv_mosquitto_broker`, `sdv_bridge_ae`, `sdv_influxdb`, and `sdv_grafana` initialize (Node-RED is decommissioned).
3. The Base CSE governance bootstrap script (`deploy_acme_fleet_cse_base.py`) executes: * Verifies CSEBase reachability (`GET /cse-in`). * Provisions `acp_fleet_manager` (). * Provisions `grp_fleet_telemetry` () with an empty member array (`mid = []`). * **Result:** Clean, production-ready Base CSE with zero registered vehicle AEs.

7.2 Scenario B: Commissioning a New Vehicle AE Instance (In-Vehicle Raspberry Pi 5)

Each physical vehicle executes its onboarding sequence independently from its onboard Raspberry Pi 5: 1. **Hardware Power-On:** ESP32 boots FreeRTOS, begins sensor sampling, and establishes local bus communication with the Raspberry Pi 5. 2. **Identity Resolution:** Raspberry Pi 5 agent reads local VIN and computes deterministic rn_{AE}. 3. **Reconcile-on-Boot Check:** * Raspberry Pi 5 issues `GET /cse-in/{{ vehicle_ae_rn }}` over WAN to the Server/VM with `X-M2M-Origin: CAdmin` (or authenticated client cert). * **Branch 1 (New Vehicle - 404 Not Found):** * Issues `POST /cse-in` with `ty=2` to create `<m2m:ae>` using deterministic rn. * Iterates through the canonical sensor manifest to create the 8 standard containers (`<m2m:cnt>`). * Attaches downlink subscription to `/commands`. * Registers its `raw-telemetry` container in Base CSE's `grp_fleet_telemetry.mid`. * **Branch 2 (Reconnecting Vehicle - 200 OK):** * Validates existing container tree. * Reclaims its assigned `aei` and immediately resumes telemetry commits. 4. **Steady-State Ingestion:** Raspberry Pi 5 posts telemetry instances (`<m2m:cin>`) via native oneM2M TS-0010 MQTT or Mca REST directly to its own `/raw-telemetry` container on the Server/VM.

7.3 Scenario C: Decommissioning / Retiring a Vehicle AE Instance

When a vehicle is decommissioned or retired from the fleet: 1. Administrative command or vehicle departure agent issues `DELETE /cse-in/{{ vehicle_ae_rn }}` against the Server/VM. 2. Base CSE cascade-deletes the vehicle's `<m2m:ae>`, all 8 child `<m2m:cnt>` containers, all `<m2m:cin>` records, and subscriptions. 3. Governance reconciliation script removes the deleted container URI from `grp_fleet_telemetry.mid`. 4. Other active vehicle AE instances running on other Raspberry Pi 5 units remain entirely unaffected.

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8. ENVIRONMENT AND CONFIGURATION CONTRACTS

8.1 Base CSE Server / VM Environment (`deploy/.env`)

The Base CSE host maintains infrastructure-wide variables:

```
# Infrastructure & Gateway Addressing
GATEWAY_IP=192.168.122.91
ONEM2M_PORT=8080
ONEM2M_CSE_ID=id-in
ONEM2M_CSE_NAME=cse-in

# Security & Governance
BASE_CSE_ADMIN_ORIGINATOR=CAdmin
FLEET_ACP_NAME=acp_fleet_manager
FLEET_GROUP_NAME=grp_fleet_telemetry

# Service Endpoints
MOSQUITTO_PORT=8883
INFLUXDB_HOST_PORT=8087
GRAFANA_PORT=3000
```

8.2 In-Vehicle Raspberry Pi 5 Environment (`vehicle/.env` per vehicle)

Each physical vehicle host maintains its local instance parameters:

```
# Vehicle Hardware & Cryptographic Identity
VEHICLE_VIN=VIN-WAUZZZ-SDV-2026-0042
VEHICLE_NODE_ID=sdv-car-0042
CLIENT_CERT_PATH=/etc/sdv/certs/client.crt
CLIENT_KEY_PATH=/etc/sdv/certs/client.key

# Target Central Base CSE Server / VM Connection
BASE_URL=http://192.168.122.91:8080
CSE_RN=cse-in
ORIGINATOR=CAdmin
APP_ID=N.org.fhtw.sdv.cabin
RVI=3

# Local In-Vehicle Instrumentation Configuration
LOCAL_ESP32_PORT=/dev/serial/by-id/usb-Espressif_ESP32-S3_001
DOWNLINK_NOTIFICATION_URL=mqtt://192.168.122.91:8883/oneM2M/req/id-in/C_ae_vin_wauzzz_0042/json
```

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9. QUALITY ATTRIBUTES AND VERIFICATION MATRIX

Quality Attribute	Requirement Specification	Verification Method
Singleton Guarantee	Exactly one Base CSE exists per network domain on the Server/VM. Zero duplicate CSEBase roots permitted.	Unit & integration test `make validate-base-cse` verifying singular `<m2m:cb>`.
AE Scalability	Fleet capacity scales to N independent vehicle AEs on distributed Raspberry Pi 5 units without code modifications.	Iterative test harness executing onboarding for M arbitrary VINs.
Collision Immunity	Two vehicles with different VINs must never claim or overwrite the same AE tree on the Base CSE.	Ingestion collision test asserting HTTP 409 / 403 on duplicate registration attempts.
Zero Initial State	Fresh deployment of Base CSE on the Server/VM must start with zero vehicular AEs.	Preflight audit confirming `len(cb.childResource[ty==2]) == 0`.
Native Ingestion	Zero intermediary proxy containers; native oneM2M TS-0010 MQTT / Mca HTTP.	Integration test verifying Node-RED absence and direct primitive routing.
Fail-Safe Containment	Network disconnection of Vehicle A's Raspberry Pi 5 has zero impact on Vehicle B's ingestion loop.	Network fault injection testing with `inject_faults.py`.